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A Framework for Rigorous Evaluation of Model-Agnostic Explainability Methods: Multi-Metric Statistical Benchmarking, Operational Protocol, and Reproducibility

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Abstract

Evaluating explainability methods requires more than a single faithfulness proxy. We present a modular benchmarking framework centered on quantitative XAI quality metrics: fidelity, stability, sparsity, computational cost, and faithfulness gap, plus an explicit method for operating the framework end-to-end. On the UCI Adult benchmark, we use a staged protocol (EXP1 calibration/reproducibility and EXP2 comparative/robustness benchmarking); the robustness cohort currently contains 250 of 300 planned configurations (83.3% coverage). Across complete model-size blocks (5 models, $N \in \{50, 100, 200\}$), Friedman tests indicate significant method differences for fidelity ($\chi^2 = 42.12$, $p = 3.78 \times 10^{-9}$), stability ($\chi^2 = 43.88$, $p = 1.60 \times 10^{-9}$), sparsity ($\chi^2 = 35.64$, $p = 8.92 \times 10^{-8}$), faithfulness gap ($\chi^2 = 45.00$, $p = 9.25 \times 10^{-10}$), and runtime ($\chi^2 = 27.72$, $p = 4.16 \times 10^{-6}$). SHAP leads on fidelity/stability, DiCE leads on sparsity, and LIME is generally fastest and most practical outside SVM-KernelSHAP bottlenecks. We release the framework, operation protocol, and artifacts with explicit data-quality caveats for reproducible benchmark use. LLM-based semantic scoring is deferred to a dedicated follow-up study.

Keywords: Explainable AI, benchmark methodology, reproducibility, statistical evaluation, model-agnostic explanations

1 Introduction

XAI evaluation remains fragmented across incompatible metrics and ad hoc setups. Most comparisons over-index on fidelity and omit stability, sparsity, and semantic interpretability. This work operationalizes a benchmark-first approach:

1. A generic trainer/explainer architecture for model-agnostic evaluation.

2. A prescriptive operation method that defines how the framework is run, audited, and reported.

3. Multi-metric quantitative scoring with explicit trade-off analysis.

4. Reproducibility and statistical rigor suitable for benchmark publication.

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2 Related Work and Framework Design

2.1 Literature-Grounded Rationale

The literature motivating this design supports five requirements:

1. Multi-metric evaluation is necessary.
2. Faithfulness is multi-faceted.
3. Computational tractability must be reported explicitly.
4. Benchmark transparency and artifact accounting are first-order requirements.
5. LLM-judge scoring should be treated as a separate calibrated track.

2.2 Architecture

The framework decouples model training, explanation generation, and evaluation through a registry/factory design. Current model coverage includes `logreg`, `rf`, `xgb`, `svm`, and `mlp`, with `lime`, `shap`, `anchors`, and `dice` explainers.

2.3 Metric Suite (Primary)

Primary quantitative metrics are:

- Fidelity
- Stability
- Sparsity
- Cost
- Faithfulness gap

2.4 Paper A Operational Outcome: Framework Operation Method (FOM-7)

Paper A contributes not only benchmark findings, but also a reusable operating method for end-to-end benchmark execution, auditing, and claim formation.

3 Methodology

3.1 Experimental Cohorts and Scope

We intentionally organize evidence into staged cohorts so that development, reliability qualification, and confirmatory inference are not conflated. This design reduces circularity (tuning and claiming on the same artifacts) and makes the evidence path for each claim auditable.

We separate evidence into four cohorts:

- **EXP1 pilot/calibration** (`exp1_adult/results`)

Table 1: Role of each evidence cohort in Paper A.

Cohort	Primary objective	Role in manuscript claims
EXP1 pilot/calibration	Verify metric behavior and tune explainer/evaluation settings before large-scale execution.	Used for implementation sanity and calibration only; not used for primary significance claims.
EXP1 reproducibility	Measure run-to-run variance under repeated seeds on core configurations.	Used for reproducibility evidence (CV and dispersion profiles).
EXP2 comparative	Provide complete method/model breadth under fixed-seed constraints.	Used as balanced comparative layer and coverage check.
EXP2 robustness	Provide multi-seed and multi-sample-size evidence for inferential testing under controlled heterogeneity.	Primary inferential cohort for Friedman/Nemenyi/Wilcoxon analyses and cross-method conclusions.

- **EXP1 reproducibility cohort** (`exp1_adult/reproducibility`)
- **EXP2 comparative** (`exp2_comparative/results`)
- **EXP2 robustness cohort** (`exp2_scaled/results`)

This EXP1/EXP2 separation is methodologically necessary for four reasons: (i) it preserves tuning-evaluation separation, (ii) it isolates implementation validity from comparative validity, (iii) it keeps artifact failures explicit instead of hidden in pooled aggregates, and (iv) it constrains claims to cohorts that are fit for each claim type.

Operationally, the structure is analogous to train/dev/test discipline, adapted to XAI system evaluation: EXP1 functions as development and reliability qualification, while EXP2 functions as comparative and inferential qualification.

Accordingly, Paper A inferential claims are driven by EXP2, while EXP1 is retained for calibration and reproducibility reporting.

3.2 Research Questions and Hypotheses

This section specifies the inferential targets and separates confirmatory discrimination questions from robustness/stability questions.

Let $\mathcal{K}$ denote explanation methods, let $\mathcal{B}$ denote evaluation blocks with $b = (\text{model}, N)$, and let $y_{k,b,m}$ denote the run-level aggregated score for method k in block b on metric m. We evaluate $m \in \{\text{fidelity}, \text{stability}, \text{sparsity}, \text{faithfulness gap}, \text{cost}\}$. These metrics represent distinct quality dimensions; therefore, single-metric dominance is not interpreted as global superiority.

RQ1 (method separation). Do explanation methods differ significantly on each metric when compared over repeated blocks? For each metric m:

$$H_{0,m}^{(1)} : \text{all methods have equal median rank across blocks,} \tag{1}$$

$$H_{1,m}^{(1)} : \text{at least one method differs in median rank.} \tag{2}$$

Rejection of $H_{0,m}^{(1)}$ indicates that the benchmark discriminates methods on metric m under observed variance and coverage.

RQ2 (trade-off stability). Are method trade-offs stable across model families and sample sizes, or context-dependent? We formalize this through two complementary hypotheses:

$$H_{0,m}^{(2a)} : \text{method ordering for metric } m \text{ is stable across blocks,} \tag{3}$$

$$H_{1,m}^{(2a)} : \text{method ordering for metric } m \text{ shifts across blocks,} \tag{4}$$

$$H_{0,m}^{(2b)} : \text{pairwise method differences on } m \text{ do not vary systematically} \tag{5}$$

$$\text{with model family or sample size,} \tag{6}$$

$$H_{1,m}^{(2b)} : \text{pairwise method differences on } m \text{ vary systematically} \tag{7}$$

$$\text{with model family and/or sample size.} \tag{8}$$

RQ2 is interpreted as a portability question: we treat context-dependent rank reversals or large cross-stratum effect shifts as evidence that trade-off guidance is not stable.

Methodological scoping in this manuscript is explicit: RQ1 is the primary confirmatory layer; RQ2 is the robustness layer connecting inferential outcomes to external validity and deployment relevance.

3.3 Dataset, Preprocessing, and Leakage Controls

The Adult Income benchmark is loaded through a dedicated pipeline (`src/data_loading/adult.py`) with explicit source fallback, schema control, and transformation traceability.

Data provenance and schema control. The loader uses OpenML Adult v2 (`data_id=1590`) as primary source. If unavailable, it falls back to direct UCI files (`adult.data`, `adult.test`). Cached artifacts are checksum-validated and stored with source and schema metadata (shape, columns, timestamp, and package versions). The enforced schema has 15 columns: one binary target (`income`), 6 numeric predictors, and 8 categorical predictors.

Cleaning includes whitespace normalization, conversion of "?" markers to `NaN`, target canonicalization (`<=50K`/`<=50K.`$\rightarrow$ 0, `>50K`/`>50K.`$\rightarrow$ 1), duplicate removal, and numeric type coercion. In the current artifact state, the cleaned corpus contains 48,790 instances.

Split discipline and deterministic execution. Data are split using a stratified holdout protocol: `train_test_split(test_size=0.2, stratify=y, random_state=seed)`. This preserves class prevalence across partitions while enabling exact reproducibility for each seed and controlled seed variation across cohorts.

Table 2: Adult dataset feature groups used in preprocessing.

Group	Variables
Numeric (6)	`age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week`
Categorical (8)	`workclass, education, marital-status, occupation, relationship, race, sex, native-country`
Target	`income` mapped to {0,1}

Preprocessing design. Transformations are implemented with a `ColumnTransformer`:

- Numeric branch: `StandardScaler` on six numeric variables.
- Categorical branch: `OneHotEncoder` on eight categorical variables.
- Encoder settings: `handle_unknown='ignore'` and `sparse_output=False`.

The encoder is fitted on training data only. Unseen categories in test instances are safely ignored, preventing transform-time failures. In the current training vocabulary, one-hot expansion yields 102 categorical columns; with 6 scaled numeric columns, the processed design matrix has 108 features.

Leakage controls and feature-space alignment. Leakage prevention is enforced by fitting transformations on training data only and reusing the fitted transformer for test and explanation pipelines. Feature names are extracted from the fitted transformer and propagated to explainers and metric engines to avoid attribution-index mismatch. In `ExperimentRunner`, when a persisted preprocessor artifact exists (`preprocessor.pkl` or `preprocessor.joblib`), it is loaded and reused so that explanation-time transformations exactly match model-training transformations.

Construct caveat. Adult is suitable for tabular XAI benchmarking, but contains socio-demographic attributes and proxies (e.g., `sex`, `race`, `native-country`). The protocol here addresses preprocessing and leakage rigor; it does not remove fairness-related construct limitations of the data.

3.4 Experimental Design and Analysis Units

3.4.1 Design Type and Inferential Target

The EXP2 comparative study is a crossed, repeated-seed factorial design with model family $g \in \mathcal{G}$, explainer $k \in \mathcal{K}$, seed $s \in \mathcal{S}$, and sampling intensity $n \in \mathcal{N}$, where

$$\mathcal{G} = \{\texttt{logreg}, \texttt{rf}, \texttt{xgb}, \texttt{svm}, \texttt{mlp}\},$$

$$\mathcal{K} = \{\texttt{lime}, \texttt{shap}, \texttt{anchors}, \texttt{dice}\},$$

$$\mathcal{S} = \{42, 123, 456, 789, 999\}, \qquad \mathcal{N} = \{50, 100, 200\}.$$

Each executed configuration yields a run-level value $y_{g,k,s,n,m}$ for metric m. Because seed coverage is partially unbalanced in realized artifacts, confirmatory inference is defined on block-level summaries over $b = (g, n)$:

$$\bar{y}_{k,b,m} = \frac{1}{|S_{k,b}|} \sum_{s \in S_{k,b}} y_{g,k,s,n,m},$$

where $S_{k,b} \subseteq \mathcal{S}$ denotes analyzable seeds for method k in block b. This prevents methods with more successful seed executions from receiving disproportionate influence in omnibus tests.

3.4.2 Planned Matrix and Realized Coverage

The planned robustness matrix contains

$$|\mathcal{G}| \times |\mathcal{K}| \times |\mathcal{S}| \times |\mathcal{N}| = 5 \times 4 \times 5 \times 3 = 300$$

run configurations (see `configs/experiments/exp2_scaled/manifest.yaml`). In the current artifact snapshot, 250 files are present in `experiments/exp2_scaled/results`, of which 233 are analyzable, 16 are empty, and 1 is malformed JSON.

For global method comparisons, we retain only block-complete model-size contexts with at least one analyzable run per method. This yields 15/15 complete blocks (5 models $\times$ 3 sample sizes). Within those blocks, seed availability is heterogeneous:

- SHAP: 2–5 seeds per block (mean ≈ 4.33)
- LIME: 5 seeds per block (mean $= 5.00$)
- Anchors: 1–5 seeds per block (mean ≈ 2.87)
- DiCE: 2–5 seeds per block (mean ≈ 3.33)

No imputation is applied for missing runs.

3.4.3 Within-Run Sampling Protocol

Each run uses stratified error-quadrant sampling (`src/evaluation/sampler.py`) over $\{\text{TP}, \text{TN}, \text{FP}, \text{FN}\}$. The configuration parameter `samples_per_class` is interpreted as target samples per quadrant, so nominal run size is $4N$.

Realized run size is lower when (i) one or more quadrants has fewer than N available test instances for a given model/seed or (ii) sampled instances fail during explainer evaluation and are dropped. Across analyzable EXP2 runs, realized run size ranges from 27 to 800 instances (median 400). Across all retained instance evaluations (103,577 total), quadrant composition remains near-balanced: TP 26,523 (25.6%), TN 26,050 (25.2%), FP 25,200 (24.3%), and FN 25,804 (24.9%).

3.4.4 Analysis Units and Anti-Pseudoreplication Strategy

The design separates instance-level logging from inferential units:

1. Instance level: raw per-case metric outputs.

2. Run level: per-configuration aggregate $y_{g,k,s,n,m}$.

3. Block level: per-method summaries $\bar{y}_{k,b,m}$ for $b = (g, n)$.

4. Matched-pair level: SHAP–LIME contrasts on shared (g, s, n) cells.

Global Friedman/Nemenyi tests are run on a 15×4 block-method matrix (15 model-size blocks, 4 methods), avoiding pseudo-replication from instance-level values and preventing seed replicates from being treated as independent blocks. For SHAP–LIME contrasts, matched analyzable sets are:

$$|\mathcal{P}_{\text{all models}}| = 65, \qquad \left|\mathcal{P}_{\texttt{logreg/rf/xgb}}\right| = 45.$$

3.4.5 Retention and Exclusion Rules

Inferential datasets are constructed under fixed artifact-qualification rules:

1. Parseable JSON artifact required (malformed files excluded).

2. Non-empty `instance_evaluations` required for run-level analysis.

3. No synthetic completion of missing runs or seeds.

4. Omnibus tests restricted to block-complete contexts.

5. Paired tests restricted to explicitly matched SHAP–LIME cells.

This makes the distinction explicit between design intent (300-run full factorial) and inferential realization (artifact-qualified subsets).

3.5 Experimental Design Protocol (Replication Specification)

This subsection specifies the executable protocol used for Paper A. The goal is to make the EXP2 pipeline reproducible without undocumented assumptions.

3.5.1 Purpose and Hypotheses Linkage

The protocol instantiates Section 3.2 questions in operational form:

1. RQ1: method separation across primary quantitative metrics.

2. RQ2: stability of method trade-offs across model family and sampling intensity.

Reliability and faithfulness are evaluated as properties of the full (model, explainer, data, metric) system.

3.5.2 Experimental Factors and Conditions

Independent variables are defined by `configs/experiments/exp2_comparative` and `configs/experiments/exp2_scaled`:

1. Dataset (fixed): Adult Income (`dataset: adult`).
2. Model family: `logreg`, `rf`, `xgb`, `svm`, `mlp`.
3. Explainer method: `lime`, `shap`, `anchors`, `dice`.
4. Seeds (robustness cohort): $\{42, 123, 456, 789, 999\}$.
5. Sampling intensity: `samples_per_class` $= N$, $N \in \{50, 100, 200\}$.
6. Stability perturbation regime: Gaussian noise with `stability_perturbations`=15 and default `stability_noise_level`=0.1.
7. Method-specific secondary flag: `counterfactual=true` for DiCE.

Planned robustness design size is $5 \times 4 \times 5 \times 3 = 300$ configurations (see Table X).

Ablation scope is explicitly fixed for Paper A:

1. Paper A is a comparative benchmark, not a component-ablation study.
2. No component-removal or hyperparameter-factor ablation enters confirmatory inference.
3. EXP1 tuning artifacts are calibration records only and are excluded from inferential sets.
4. Ablation-driven causal attribution is deferred to a separate extension study (Paper B).

3.5.3 Controls and Baselines

Within each matched cell $(\text{model}, \text{seed}, N)$, the following are held constant:

1. Black-box model artifact path.
2. Preprocessor artifact and transformed feature space.
3. Split and sampling seeds.
4. Evaluation sampler and metric engine code path.
5. Primary metric set and run-level aggregation rules.

Comparator set is `lime/shap/anchors/dice`. Sanity-control baselines (random attribution, label randomization, parameter randomization) are not yet included. Therefore, Paper A claims are restricted to relative method performance under trained-model conditions; no claim is made about explanation validity under label/model randomization perturbations.

3.5.4 Data Protocol

Data handling follows Section 3.3 and `src/data_loading/adult.py`:

1. Adult cleaning: whitespace normalization, missing-value canonicalization, target canonicalization, duplicate removal.
2. Stratified split: `train_test_split(test_size=0.2, stratify=y, random_state=seed)`.
3. Preprocessor fit on training split only (unless persisted preprocessor is provided).
4. Shared train/test transformation and feature-name propagation.
5. TP/TN/FP/FN stratified evaluation sampling with target N per quadrant.

For seed 42, split sizes are train 39,032 and test 9,758 ($y = 0$: 29,687/7,422; $y = 1$: 9,345/2,336). Nominal run size is $4N$, with lower realized size when quadrant availability is below N.

3.5.5 Model Training Protocol

EXP2 loads frozen EXP1 model artifacts; models are not retrained during EXP2. Binary target objective is income classification.

Persisted model configurations:

1. LogReg: `solver=lbfgs`, `max_iter=1000`, `random_state=42`.
2. RF: `n_estimators=50`, `max_depth=15`, `min_samples_leaf=1`, `class_weight=balanced_subsample`, `random_state=42`.
3. XGB: `n_estimators=100`, `max_depth=6`, `learning_rate=0.1`, `objective=binary:logistic`, `eval_metric=logloss`, `random_state=42`.
4. SVM: `kernel=rbf`, `C=1.0`, `gamma=scale`, `probability=true`, `random_state=42`.
5. MLP: `hidden_layer_sizes=[100]`, `activation=relu`, `solver=adam`, `max_iter=500`, `random_state=42`.

Early stopping is implemented only in `XGBoostTrainer` when validation data are passed; it is not active in the released EXP2 model artifacts. No post-hoc probability calibration is applied. Calibration interpretation policy:

1. All methods are evaluated on identical uncalibrated model outputs per matched cell.
2. Inferential statements are comparative (ranking/trade-off), not calibration claims.
3. No calibration-dependent claim is made in Paper A.

3.5.6 Explanation Generation Protocol

All explainers are called through `ExplainerWrapper.explain_instance` and timed per instance.

SHAP. `tree` mode for RF/XGB and `kernel` mode for SVM/MLP/LogReg, `n_background_samples=50`. Tree mode uses interventional perturbation and probability output; class-1 attributions are retained.

LIME. `num_samples=1000`, `num_features=10`, `kernel_width=3.0`, `discretize_continuous=False`; class-1 explanation map is converted to dense vector.

Anchors. `AnchorTabular` fitted on transformed training data; rule threshold 0.95; anchor membership mapped to binary feature-importance vector.

DiCE. Backend method `random`; `desired_class="opposite"`; importance proxy is absolute original–counterfactual feature difference.

Implementation caveats:

1. Anchors threshold is hardcoded at 0.95 in wrapper call.
2. DiCE YAML `num_counterfactuals` is not propagated; runtime uses `total_CFs=1`.

Parameter-binding policy for reported runs:

1. SHAP: `explainer_type` and `n_background_samples` from YAML are applied.
2. LIME: `kernel_width` from YAML params is applied; `num_samples` and `num_features` are bound through schema fields (defaults 1000/10), with duplicate `params` entries dropped.
3. Anchors: YAML `threshold` is not consumed; effective threshold is fixed at 0.95.
4. DiCE: YAML `num_counterfactuals` is not consumed; effective setting is `total_CFs=1` with `method=random`.

These bindings define the effective treatment conditions for EXP2 claims.

Runtime constraints: explanation cost is measured with `perf_counter`; no explicit timeout/memory cap in YAML; batch runner enforces one evaluation worker per experiment process to prevent nested parallelism explosion. Run-level cost profile over analyzable EXP2 runs (`outputs/analysis/paper_a_exp2_stats/exp2_run_level_metrics.csv`):

Method	Runs	Mean (ms)	Median (ms)	P95 (ms)	Max (ms)
SHAP	65	503870.23	1098.11	1603727.03	9393990.21
LIME	75	3660.68	65.73	13192.05	85825.65
Anchors	43	28762.43	34063.36	64568.75	67997.67
DiCE	50	15337.17	9759.79	34172.81	37164.19

3.5.7 Evaluation Metrics and Measurement Procedure

Metrics are computed per instance in `MetricsEngine`:

1. Cost: explanation time in milliseconds.
2. Sparsity: active-feature ratio ($|w_i| > \tau$, $\tau = 10^{-4}$).

3. Fidelity proxy: faithfulness correlation score.

4. Faithfulness gap: prediction shift after masking top-k features ($k = 5$).

5. Stability: mean pairwise cosine similarity under Gaussian perturbations.

Faithfulness implementation (`src/metrics/faithfulness.py`):

$$\Delta_k = \left| f(x) - f\left(x_{\text{mask-top-}k}\right) \right|, \tag{9}$$

$$\rho = \operatorname{corr}\left(|w_i|, |f(x) - f(x_{-i})|\right). \tag{10}$$

Pipeline mapping is `fidelity` $\leftarrow \rho$ and `faithfulness_gap` $\leftarrow \Delta_k$.

Stability implementation (`src/metrics/stability.py`):

$$x^{(t)} = x + \epsilon^{(t)}, \qquad \epsilon^{(t)} \sim \mathcal{N}(0, \sigma^2), \tag{11}$$

$$S = \frac{2}{T(T-1)} \sum_{a<b} \cos\left(e^{(a)}, e^{(b)}\right). \tag{12}$$

Secondary track: `cf_sensitivity` is computed when `counterfactual=true`; domain-alignment metric exists but is disabled in EXP2 configs (`domain=false`).

Aggregation is instance $\rightarrow$ run (`mean/std/min/max/count`) $\rightarrow$ block summary over (model, N).

EEU note: Explanation Energy Units are specified in metric-design artifacts but not computed in current EXP2 runtime. EEU is therefore excluded from confirmatory inference, and Paper A cost claims are defined strictly on wall-clock milliseconds.

3.5.8 Statistical Analysis Plan

Confirmatory dataset construction follows artifact qualification rules (Sections 3.4.5 and 3.6). Planned robustness grid is 300 runs; current snapshot has 250 present and 233 analyzable runs.

Inferential procedures:

1. Friedman test per metric over methods with (model, N) blocks.

2. Nemenyi post-hoc pairwise test when Friedman is significant.

3. Paired Wilcoxon SHAP-vs-LIME on matched $(\text{model}, \text{seed}, N)$ cells.

Nemenyi provides family-wise control for pairwise localization. Across-metric multiplicity is controlled with Holm-Bonferroni adjustment over the five primary metrics, applied separately to the Friedman family and to each Wilcoxon family (45-pair and 65-pair matched sets). Uncertainty is reported with mean/std/CV and, where generated, t-based and bootstrap CIs.

Effect-size reporting is exported with inference artifacts:

1. Friedman layer: Kendall's W for each metric.

2. Paired SHAP–LIME layer: Cohen's d_z and median paired differences.

3.5.9 Robustness and Sensitivity Analyses

Implemented robustness axes:

1. Seed heterogeneity (5 seeds).
2. Sample-size heterogeneity ($N \in \{50, 100, 200\}$ per quadrant).
3. Perturbation stability under Gaussian noise ($T = 15$, $\sigma = 0.1$ default).

Additional reproducibility sensitivity is reported from the EXP1 repeated-seed cohort (`reproducibility_report.csv`).

Not included in Paper A core protocol: adversarial perturbation campaigns and causal-intervention benchmarks beyond proxy counterfactual sensitivity. Robustness interpretation in Paper A is limited to seed, sample-size, and Gaussian-noise perturbation dimensions. Reporting rule for robustness consistency:

1. SHAP–LIME median-difference sign must match between 45-cell and 65-cell matched analyses.
2. Holm-adjusted significance decision must not flip between these analyses.

Metrics violating either condition are reported as sensitivity-unstable.

3.5.10 Reproducibility Package and Rerun Instructions

Release package currently includes source code, experiment YAMLs, model and preprocessor artifacts, per-run JSON/CSV outputs, batch manifest, and figure generation scripts.

Minimal rerun sequence:

1. Install dependencies from `requirements-frozen.txt`.
2. Verify or regenerate `experiments/exp1_adult/models` artifacts.
3. Run EXP2 robustness batch: `python3 scripts/run_batch_experiments.py --config-dir configs/experiments/exp2_scaled --output outputs/batch_results.csv --parallel`.
4. Run coverage audit: `python3 scripts/check_missing_results.py`.
5. Generate summaries and figures.

Execution provenance anchor for this manuscript snapshot:

1. Reference regeneration host: Apple M3 Pro, macOS 26.3 (arm64), 11 logical CPUs, 18 GB RAM, Python 3.13.2.
2. Dependency lock hash: `sha256(requirements-frozen.txt)` = `b52d5a1f2e2edd5ada372ca66d18ef1447712fdd2c21f004d7e1f46a5ef9c6dc`.
3. Immutable DOI is not yet minted in this repository state; current provenance key is `outputs/batch_manifest.json` with git hash `9fc70eb1e218a11f2fdd4bdb3aab3ea10a799a6f`.

3.6 Statistical Analysis Pipeline (Replication-Grade Specification)

This subsection defines the executable protocol used to produce all Paper A inferential artifacts from repository-tracked inputs (see Table X and Appendix X).

3.6.1 Experimental Factors and Conditions

Independent variables are instantiated from `configs/experiments/exp2_comparative`, `configs/experiments/exp2_scaled`, and `configs/experiments/exp2_scaled/manifest.yaml`:

1. Dataset (fixed): `adult`.
2. Model family: `logreg`, `rf`, `xgb`, `svm`, `mlp`.
3. Explainer method: `shap`, `lime`, `anchors`, `dice`.
4. Seeds: $\{42, 123, 456, 789, 999\}$.
5. Sampling intensity: `samples_per_class`=N, $N \in \{50, 100, 200\}$ (robustness cohort).
6. Stability perturbation regime: `stability_perturbations`=15 and default `stability_noise_level`=0.1.
7. Counterfactual metric activation: `counterfactual=true` only for DiCE runs.

Planned robustness grid size is $5 \times 4 \times 5 \times 3 = 300$. No component-removal ablation is included in Paper A confirmatory inference.

3.6.2 Controls and Baselines

Within each matched $(\text{model}, \text{seed}, N)$ cell, the following are held constant:

1. Model artifact path.
2. Preprocessor artifact and transformed feature space.
3. Split/sampling random-state values.
4. Evaluation sampler and metric-engine implementation.
5. Metric set and aggregation logic.

Baselines are cross-method (`shap/lime/anchors/dice`) under shared conditions. Random-attribution, label-randomization, and model-randomization sanity controls are not part of Paper A runs; therefore claims are bounded to trained-model comparative performance.

3.6.3 Data Protocol

Data handling is implemented in `src/data_loading/adult.py` and `src/evaluation/sampler.py`:

1. Adult cleaning (whitespace normalization, missing-value canonicalization, duplicate removal, target normalization).
2. Stratified split with `test_size=0.2`, `stratify=y`, `random_state=seed`.

3. Preprocessor fit on train only.
4. Shared train/test transformation and feature-name propagation.
5. TP/TN/FP/FN stratified sampling with target size N per stratum.

For seed 42, split sizes are train 39,032 and test 9,758 ($y = 0$: 29,687/7,422; $y = 1$: 9,345/2,336). Nominal run size is $4N$; realized size may be lower under stratum-capacity constraints.

3.6.4 Model Training Protocol

EXP2 evaluates frozen EXP1 model artifacts; no retraining occurs in EXP2. Persisted model settings are:

1. LogReg: `solver=lbfgs`, `max_iter=1000`, `random_state=42`.
2. RF: `n_estimators=50`, `max_depth=15`, `min_samples_leaf=1`, `class_weight=balanced_subsample`, `random_state=42`.
3. XGB: `n_estimators=100`, `max_depth=6`, `learning_rate=0.1`, `objective=binary:logistic`, `eval_metric=logloss`, `random_state=42`.
4. SVM: `kernel=rbf`, `C=1.0`, `gamma=scale`, `probability=true`, `random_state=42`.
5. MLP: `hidden_layer_sizes=[100]`, `activation=relu`, `solver=adam`, `max_iter=500`, `random_state=42`.

Early stopping is implemented only for XGBoost when validation data are passed; it is inactive in persisted EXP2 artifacts. No post-hoc probability calibration is applied.

3.6.5 Explanation Generation Protocol

Explanations are generated via `ExplainerWrapper.explain_instance` on the dense transformed feature vector (108 dimensions in current Adult vocabulary). Per-instance runtime is recorded via wall-clock timing.

1. SHAP: `tree` mode for RF/XGB, `kernel` for LogReg/SVM/MLP; `n_background_samples=50`; class-1 attributions retained.
2. LIME: `num_samples=1000`, `num_features=10`, `kernel_width=3.0`, `discretize_continuous=False`.
3. Anchors: `AnchorTabular` on transformed train matrix; effective precision threshold fixed at 0.95.
4. DiCE: backend `method=random`, `desired_class=opposite`, absolute original–counterfactual deltas as importance proxy.

Effective parameter binding caveats:

1. Anchors YAML `threshold` is not consumed; runtime threshold is fixed at 0.95.
2. DiCE YAML `num_counterfactuals` is not consumed; runtime uses `total_CFs=1`.

No explicit timeout/memory cap is set in EXP2 YAML; batch execution enforces one evaluation worker per experiment process.

3.6.6 Evaluation Metrics and Measurement Procedure

Primary metrics are computed per instance in `src/experiment/metrics_engine.py`:

1. `cost` (milliseconds),
2. `sparsity` (active-feature ratio, threshold $\tau = 10^{-4}$),
3. `fidelity` (faithfulness correlation score),
4. `faithfulness_gap` (top-k masking gap, $k = 5$),
5. `stability` (mean pairwise cosine similarity under perturbations).

Faithfulness implementation (`src/metrics/faithfulness.py`):

$$\Delta_k = \left| f(x) - f\left(x_{\text{mask-top-}k}\right) \right|, \tag{13}$$

$$\rho = \operatorname{corr}\left(|w_i|, |f(x) - f(x_{-i})|\right). \tag{14}$$

Mapping in pipeline: `fidelity`$\leftarrow \rho$ and `faithfulness_gap`$\leftarrow \Delta_k$.

Stability implementation (`src/metrics/stability.py`):

$$x^{(t)} = x + \epsilon^{(t)}, \qquad \epsilon^{(t)} \sim \mathcal{N}(0, \sigma^2), \tag{15}$$

$$S = \frac{2}{T(T-1)} \sum_{a<b} \cos\left(e^{(a)}, e^{(b)}\right). \tag{16}$$

Secondary metrics: `cf_sensitivity` only when `counterfactual=true`; domain alignment implemented but disabled in EXP2. Aggregation hierarchy is instance $\rightarrow$ run $\rightarrow$ block.

EEU is specified in EXP1 metric-design configs but not computable in current EXP2 runtime due missing memory/model-call telemetry. `[TO FILL: implement EEU instrumentation or keep EEU out of Paper A claims]`.

3.6.7 Statistical Analysis Plan

Artifact-qualified run inventory is:

1. planned runs: 300,
2. present `results.json`: 250,
3. analyzable runs: 233,
4. excluded artifacts: 16 empty and 1 malformed JSON.

Coverage imbalance is shown in Figure 1.

Global inference is restricted to 15 complete (model, N) blocks. SHAP–LIME matched sets are 45 cells (`logreg/rf/xgb`) and 65 cells (all models). Inferential stack (`scripts/run_exp2_statistical_analysis.py`):

1. Friedman omnibus test per metric.

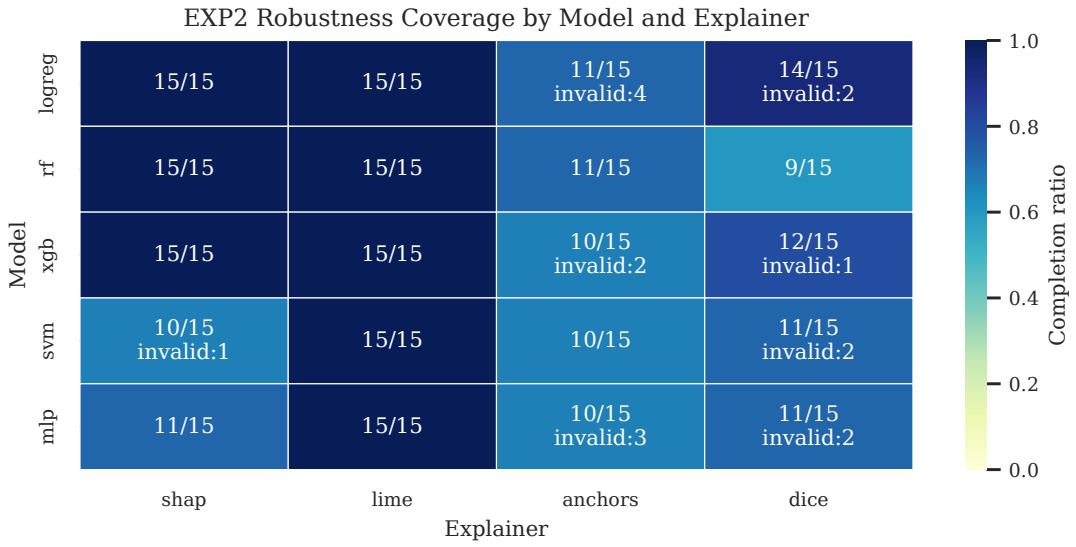

Figure 1: Coverage map of the EXP2 robustness grid by model and explainer. Cell text reports files present out of 15 expected (5 seeds × 3 sample sizes), with invalid artifact counts where applicable.

2. Nemenyi post-hoc localization per metric.

3. Two-sided Wilcoxon signed-rank SHAP–LIME tests on matched cells.

Multiplicity control: Nemenyi within metric; Holm-Bonferroni across the five primary metrics for each inferential family (Friedman, Wilcoxon-45, Wilcoxon-65). Uncertainty/effect reporting: mean, std, CV, t and BCa bootstrap CIs, Kendall's W, Cohen's d_z, and median paired differences.

3.6.8 Robustness and Sensitivity Analyses

Implemented robustness axes are seed heterogeneity (5 seeds), sampling-intensity heterogeneity ($N = 50/100/200$), and Gaussian perturbation stability. Sensitivity controls:

1. Re-estimate SHAP–LIME contrasts on both 45-cell and 65-cell matched sets.

2. Restrict global inference to block-complete contexts.

3. Exclude malformed/empty artifacts without reconstruction.

Robustness claim rule:

1. SHAP–LIME median-difference sign must match between 45-cell and 65-cell analyses.

2. Holm-adjusted significance decision must not flip between the two analyses.

Adversarial stress testing is not part of Paper A. `[TO FILL: adversarial stress protocol with pre-registered deviation thresholds]`.

3.6.9 Reproducibility Package

Released package includes code, experiment YAMLs, frozen model/preprocessor artifacts, run outputs, aggregate outputs, inferential exports, and figure generation scripts. Reference provenance anchors:

1. Host used for manuscript regeneration: Apple M3 Pro, macOS 26.3 (arm64), 11 logical CPUs, 18 GB RAM, Python 3.13.2.

2. `sha256(requirements-frozen.txt)` = `b52d5a1f2e2edd5ada372ca66d18ef1447712fdd2c21f004d7e1f46a`

3. `outputs/batch_manifest.json` git hash `9fc70eb1e218a11f2fdd4bdb3aab3ea10a799a6f`.

Minimal rerun sequence:

1. `python3 scripts/run_batch_experiments.py --config-dir configs/experiments/exp2_scaled --output outputs/batch_results.csv --parallel`

2. `python3 scripts/check_missing_results.py`

3. `python3 scripts/quick_summary.py`

4. `python3 scripts/full_summary.py`

5. `python3 scripts/generate_paper_a_figures.py`

6. `python3 scripts/run_exp2_statistical_analysis.py`

Archival status: `[TO FILL: immutable DOI bundle for camera-ready reproducibility package]`.

3.7 Inferential Implementation Details

This subsection defines the inferential workflow for EXP2 and its reproducibility support cohorts.

3.7.1 Inferential Objective and RQ Linkage

The statistical design implements Section 3.2:

1. RQ1 (method separation): test whether methods differ for each metric.

2. RQ2 (trade-off stability): test whether method contrasts remain stable across model families and sampling intensities.

All tests are executed on run/block-level units (Section 3.4), not on pooled instance rows.

3.7.2 Analysis Populations and Eligibility

Inferential inputs are artifact-qualified EXP2 outputs:

1. Planned robustness grid: 300 runs.
2. Present artifacts: 250 files.
3. Analyzable artifacts: 233 (non-empty `instance_evaluations`).
4. Invalid artifacts: 1 malformed JSON and 16 empty files.

Eligibility rules follow Section 3.4.5: parseable JSON, non-empty run metrics, no imputation, and block-complete filtering for omnibus tests.

Resulting inferential sets are:

1. Global omnibus set: 15 complete (model, N) blocks $\times$ 4 methods.
2. Primary SHAP–LIME paired set: 45 matched cells (`logreg/rf/xgb`).
3. Sensitivity SHAP–LIME paired set: 65 matched cells (all model families).
4. EXP1 reproducibility cohort: 9-seed repeated runs per core configuration.

3.7.3 Estimands, Aggregation, and Metric Directionality

For metric m, run-level estimands are $y_{g,k,s,n,m}$ and block summaries are:

$$\bar{y}_{k,b,m} = \frac{1}{|S_{k,b}|} \sum_{s \in S_{k,b}} y_{g,k,s,n,m}, \quad b = (g, n).$$

Primary metrics are fidelity, stability, sparsity, faithfulness gap, and cost. Directionality is fixed pre-analysis:

1. Higher is better: fidelity, stability, faithfulness gap.
2. Lower is better: sparsity, cost.

Directionality is used for interpretation and normalized visual summaries (see Figure X), while inferential tests operate on observed values per block.

3.7.4 Global Omnibus Test (Friedman)

For each metric:

1. Build block-by-method matrix (rows: complete (model, N) blocks; columns: methods).
2. Apply Friedman test with methods as treatments and blocks as matched contexts.
3. Evaluate significance at $\alpha = 0.05$ (Table X).

Friedman is used because the design is repeated-measures and metric distributions are not assumed Gaussian.

3.7.5 Post-hoc Localization (Nemenyi)

If Friedman rejects the null for metric m:

1. Run Nemenyi pairwise post-hoc comparisons on the same block matrix.
2. Report pairwise adjusted p-values for all method pairs.
3. Use $\alpha = 0.05$ for within-metric family-wise control.

Across-metric multiplicity is controlled via Holm-Bonferroni adjustment over the five primary metrics, applied separately to each inferential family.

3.7.6 Focused SHAP–LIME Paired Test (Wilcoxon)

For each primary metric:

1. Construct matched SHAP/LIME pairs on identical $(\text{model}, \text{seed}, N)$ cells.
2. Apply two-sided Wilcoxon signed-rank test.
3. Report paired descriptive summaries and p-values.

Primary analysis uses the 45-cell restricted set; 65-cell all-model analysis is used as sensitivity support.

3.7.7 Uncertainty Quantification

Reported uncertainty statistics include mean, standard deviation, and CV. Where configured, confidence intervals are computed via `src/analysis/confidence.py`:

1. t-distribution CIs (`compute_t_ci`).
2. Bootstrap CIs (`compute_bootstrap_ci`), BCa method with 10,000 resamples and seed 42; percentile fallback on failure.

3.7.8 Effect Size and Practical Significance

Effect-size outputs are generated alongside p-values:

1. Kendall's W for Friedman tests.
2. Cohen's d_z for paired Wilcoxon SHAP–LIME contrasts.
3. Median paired differences for location-scale interpretation.

These are exported by the EXP2 statistical driver for direct appendix use.

3.7.9 Sensitivity and Missingness Controls

Sensitivity controls in the analysis layer:

1. Re-estimate SHAP–LIME contrasts on restricted (45) and all-model (65) matched sets.
2. Restrict global inference to block-complete contexts.
3. Exclude malformed/empty artifacts without synthetic reconstruction.

Dispersion is quantified via std/CV/CI. No threshold-based practical-equivalence rule is enforced in the current protocol; robustness interpretation therefore relies on direction consistency, effect-size magnitude, and uncertainty bounds.

3.7.10 Reproducible Analysis Execution

Current reproducible analysis stack:

1. Batch execution and artifact generation.
2. Missing/malformed audit (`scripts/check_missing_results.py`).
3. Summary tables (`scripts/quick_summary.py`, `scripts/full_summary.py`).
4. Figure regeneration (`scripts/generate_paper_a_figures.py`).
5. Deterministic EXP2 inference export: `python3 scripts/run_exp2_statistical_analysis.py`.

The EXP2 driver materializes Friedman, Nemenyi, Wilcoxon, multiplicity-adjusted p-values, effect-size tables, and matched-cell inventories in `outputs/analysis/paper_a_exp2_stats/`.

3.8 Reproducibility Protocol

3.8.1 Scope and Methodological Role

This protocol defines how Section 4 quantitative claims are regenerated from repository artifacts. It links Sections 3.6–3.7 to a deterministic execution contract for RQ1/RQ2 (see Table X).

3.8.2 Definitions, Notation, and Assumptions

Run key is $r = (g, k, s, n)$ with model family g, explainer k, seed s, and sampling intensity n. Reproducibility state is decomposed into:

1. $\mathcal{C}$: configuration state (YAMLs, seed grid, CLI invocations),
2. $\mathcal{A}$: artifact state (`results.json`, `metrics.csv`, aggregate exports),
3. $\mathcal{I}$: inferential state (`outputs/analysis/paper_a_exp2_stats/`),
4. $\mathcal{V}$: variance state (`reproducibility_report.csv`).

Assumptions: EXP1 model artifacts are frozen during EXP2; config files are immutable during one run; only artifact-qualified runs enter inference.

3.8.3 Executable Reproducibility Procedure

1. Freeze protocol anchors:
   (a) config roots (`configs/experiments/exp2_scaled`, `configs/experiments/exp2_comparative`),
   (b) dependency hash: `sha256(requirements-frozen.txt)`= `b52d5a1f2e2edd5ada372ca66d18ef1447712`
   (c) provenance hash in `outputs/batch_manifest.json`.
2. Execute EXP2 batch generation.
3. Audit missing/malformed/empty artifacts.
4. Regenerate deterministic inference exports via `scripts/run_exp2_statistical_analysis.py`.
5. Regenerate summaries/figures and repeated-run reproducibility outputs.

3.8.4 Quality Checks and Acceptance Criteria

A reproduction pass is claim-eligible only if:

1. seed/model/explainer grids match declared manifests;
2. malformed and empty artifacts are explicitly enumerated;
3. omnibus inference uses only complete (model, N) blocks;
4. Section 4 claims map to versioned files under analysis/figure outputs;
5. repeated-run variance diagnostics are present.

3.8.5 Environment and Compute Disclosure

Reference regeneration host: Apple M3 Pro, macOS 26.3 (arm64), 11 logical CPUs, 18 GB RAM, Python 3.13.2. This is a provenance anchor, not an execution requirement. `[TO FILL: multi-hardware normalization protocol if runtime claims are generalized across platforms]`.

3.9 Validity Threats and Mitigations

3.9.1 Scope

Threat analysis is organized by validity class to separate implemented controls from residual uncertainties.

3.9.2 Internal Validity

1. Threat: incomplete EXP2 execution coverage (300 planned, 250 present, 233 analyzable).
2. Mitigation: explicit artifact audit, block-complete filtering, matched-cell filtering.
3. Residual risk: potential non-random missingness by method–model cell.

3.9.3 Construct Validity

1. Threat: ambiguity in metric naming/interpretation.
2. Mitigation: code-level operationalization and explicit mapping (`fidelity`←faithfulness correlation, `faithfulness_gap`←top-k masking shift).
3. Residual risk: EEU not computed in current EXP2 runtime; cost is wall-clock only.

3.9.4 Statistical Conclusion Validity

1. Threat: inflated Type-I error and pseudo-replication.
2. Mitigation: run/block analysis units, Friedman+Nemenyi+Wilcoxon stack, Holm adjustments.
3. Residual risk: no pre-registered practical-equivalence threshold.

3.9.5 External Validity

1. Threat: single benchmark dataset/task family.
2. Mitigation: diversity across five model families, four explainers, five seeds, and three sample sizes.
3. Residual risk: generalization to text/vision/time-series is not established in Paper A.

3.9.6 Implementation Validity

1. Threat: YAML-exposed parameters not fully bound at runtime (Anchors threshold, DiCE CF count).
2. Mitigation: explicit disclosure of effective runtime settings.
3. Residual risk: harmonization could shift absolute method levels.

`[TO FILL: camera-ready harmonization diff report between YAML-exposed and runtime-bound parameters]`.

3.10 Framework Operation Method (FOM-7): Executable Protocol

3.10.1 Formal Definition and Role

FOM-7 is the operational method contribution of Paper A. It is defined as:

$$\mathcal{M}_{\text{FOM7}} = \{(S_i, A_i, G_i)\}_{i=1}^{7},$$

with stage S_i, required artifacts A_i, and gate condition G_i. Progression is allowed only if $G_i = \text{pass}$.

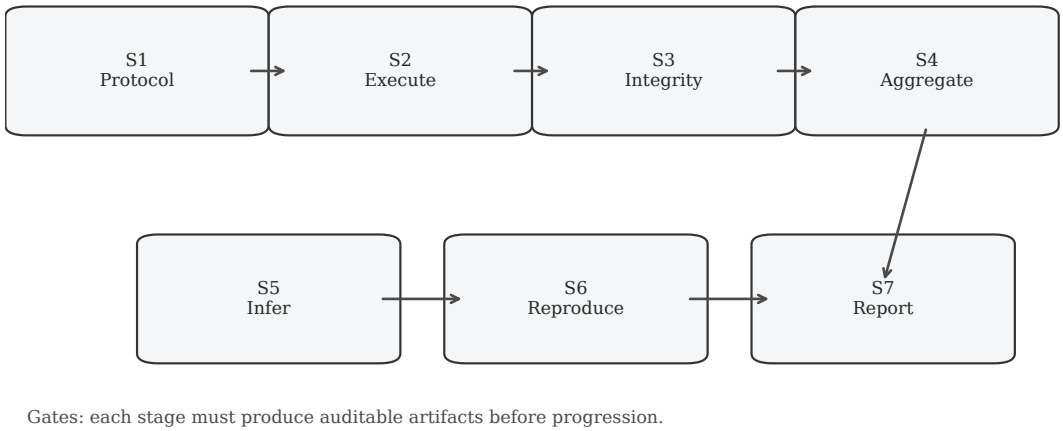

Figure 2: Operational view of FOM-7. The workflow is stage-gated: each stage must produce auditable artifacts before progression to the next stage.

3.10.2 Stage Protocol and Gate Logic

Algorithm 1 (FOM-7) maps protocol execution to stage-gated evidence production:

1. **S1 Protocol specification**: freeze study design and enumerate the run grid.
2. **S2 Controlled execution**: run batch jobs and persist per-run artifacts plus execution manifest.
3. **S3 Integrity audit**: enumerate missing, empty, and malformed artifacts.
4. **S4 Harmonization**: aggregate retained outputs into analysis-ready metric tables.
5. **S5 Statistical inference**: run Friedman, Nemenyi, and paired Wilcoxon tests.
6. **S6 Reproducibility characterization**: estimate cross-seed dispersion and CV.
7. **S7 Claim-ready reporting**: map manuscript claims to versioned artifacts and caveat ledger.

Progression from stage S_i to stage S_{i+1} is allowed only when the corresponding gate G_i is satisfied.

Figure 2 provides the operational view of FOM-7.

3.10.3 Reference Implementation Procedure

1. `python3 scripts/run_batch_experiments.py`
   `--config-dir configs/experiments/exp2_scaled`
   `--output outputs/batch_results.csv --parallel`

2. `python3 scripts/check_missing_results.py`

3. `python3 scripts/quick_summary.py`

4. `python3 scripts/full_summary.py`

5.
```
python3 scripts/run_reproducibility_study.py
  --config-dir configs/experiments --pattern "exp1_adult_*.yaml"
  --output-dir experiments/exp1_adult/reproducibility
```

6. `python3 scripts/generate_paper_a_figures.py`

7. `python3 scripts/run_exp2_statistical_analysis.py`

If any gate fails, downstream inferential outputs are not claim-eligible and the affected comparison is downgraded to descriptive-only reporting.

4 Results

4.1 Global Trade-offs (EXP2 robustness)

Figure 3 provides a direction-normalized profile view. Table 3 reports absolute means over 15 complete model-size blocks.

Table 3: Method-level means over complete model-size blocks.

Method	Fidelity	Stability	Sparsity	Faithfulness Gap	Time (ms)
SHAP	0.8176	0.7377	0.2264	0.4474	685220.23
LIME	0.5602	0.0144	0.0846	0.3342	3660.68
Anchors	0.3853	0.0006	0.0928	0.2382	25326.92
DiCE	0.1716	0.3602	0.0164	0.0988	16306.50

Figure 4 isolates the fidelity–cost frontier and its dispersion across complete blocks.

4.2 Statistical Significance

Table 4: Friedman tests across methods.

Metric	Statistic	p-value	Significant
Fidelity	42.12	3.78e-09	Yes
Stability	43.88	1.60e-09	Yes
Sparsity	35.64	8.92e-08	Yes
Faithfulness Gap	45.00	9.25e-10	Yes
Cost	27.72	4.16e-06	Yes

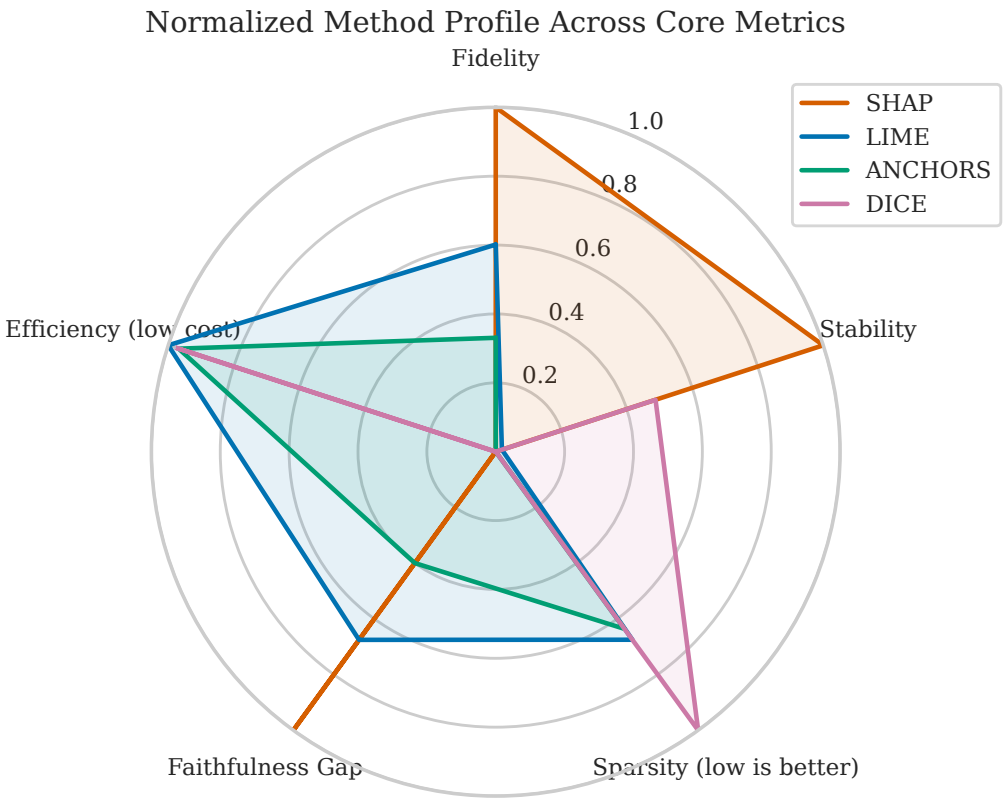

Figure 3: Normalized method profile across core metrics (higher is better after direction-aware normalization). SHAP dominates fidelity/stability, while LIME and DiCE dominate efficiency/sparsity dimensions, respectively.

Table 5: Paired Wilcoxon comparison on 45 matched configurations (`logreg/rf/xgb`).

Metric	SHAP Mean	LIME Mean	Wilcoxon p-value
Fidelity	0.8112	0.5556	5.68e-14
Stability	0.8013	0.0154	5.68e-14
Sparsity	0.3156	0.0845	5.68e-14
Faithfulness Gap	0.3924	0.3408	5.68e-14
Cost (ms)	1258.72	210.45	2.29e-06

4.3 Focused SHAP vs LIME Comparison

4.4 Reproducibility Snapshot (EXP1 repeated runs)

Fidelity CV ≤ 0.0634, stability CV ≤ 0.0826, sparsity CV ≤ 0.0441, and faithfulness-gap CV ≤ 0.0358 across baseline configurations. Cost CV reaches 0.225 for SHAP variants.

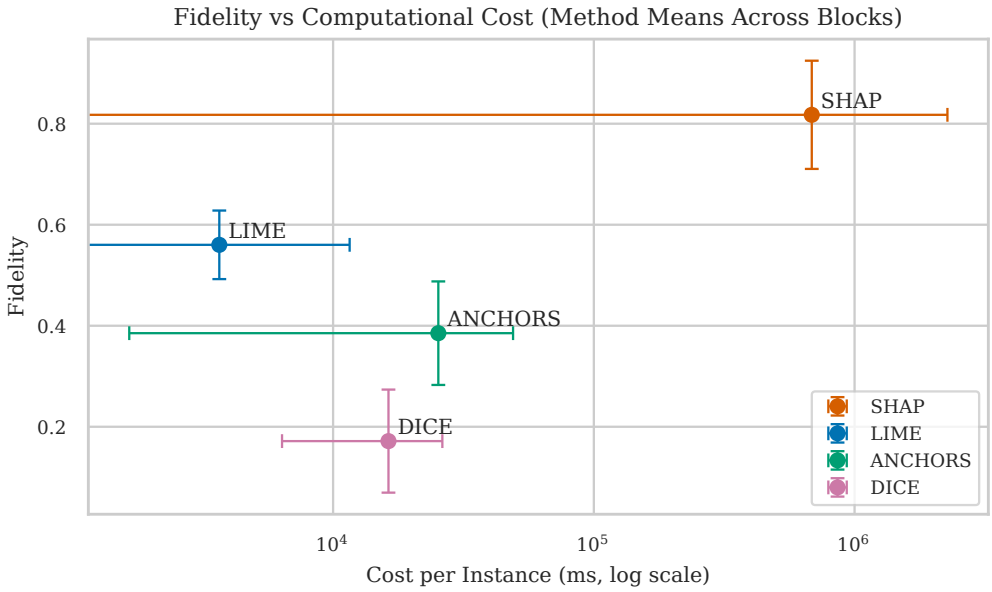

Figure 4: Fidelity–cost trade-off frontier using method means over complete blocks (error bars: block-level standard deviation). The x-axis is logarithmic due strong runtime heterogeneity across methods.

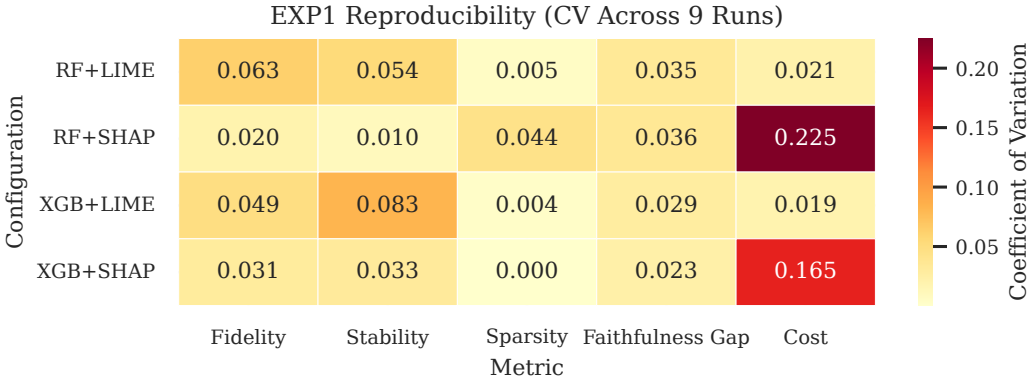

Figure 5: Reproducibility heatmap for EXP1 repeated runs (9 seeds per core configuration). Lower CV indicates stronger run-to-run stability.

Figure 5 highlights that cost variance dominates quality-metric variance in repeated runs.

5 Validity and Reporting Caveats

1. EXP2 robustness is incomplete (50 missing runs) with one malformed result file.

2. Runtime comparisons are sensitive to implementation details and hardware.
3. Semantic evaluation is out of scope for Paper A.

6 JMLR Track Positioning

1. Reproducible benchmark framework with artifact lineage and failure accounting.
2. Multi-metric protocol with non-parametric inferential testing.
3. Reusable operation method (FOM-7) for benchmark execution and audit standardization.
4. Clearly scoped quantitative benchmark; semantic track deferred.

7 Conclusion

The evidence supports a clear trade-off frontier: SHAP leads fidelity/stability, LIME leads runtime practicality, and DiCE leads sparsity. In addition to empirical findings, Paper A contributes FOM-7 as a concrete operating protocol for running, auditing, and reporting XAI benchmark studies.

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